rag -- 2026-10-24

4 new paper(s) · topic: Retrieval-augmented generation: retrieval quality, grounding and faithfulness, evaluation benchmarks, and RAG system design.

Worth reading in full

[P3] Automated Enrichment Analysis with GraphRAG: Efficient Multi-Database ORA Extraction and Biological Interpretation -- GBOR-RAG combines a novel graph-based retrieval mechanism with LLM generation and offers the most comprehensive analysis of retrieval quality, grounding, and faithfulness, making it the most informative read for the stated topic.

Themes

Several papers examine how retrieval can improve factual grounding in generation. Both the clinical-summary study and the biological-interpretation system report that adding a retrieval component reduces a faithfulness error metric (Mfaith drops from 0.242 to 0.178 in the clinical setting and claim-level faithfulness rises to 0.90 in the enrichment system) [P1][P3]. However, the clinical work notes a trade-off with readability, whereas the GraphRAG paper emphasizes faster query times and more integrated evidence subgraphs, highlighting different design priorities for retrieval-augmented generation.

The methods for integrating retrieved material also diverge. The health-summary paper uses a simple "RAG-Constrained" prompt that pulls lay-term definitions from a static README dataset, while the GraphRAG approach builds a term-gene knowledge graph and extracts a compact evidence subgraph via prize-collecting retrieval before conditioning the LLM, representing a more structured retrieval pipeline [P1][P3].

Beyond grounding, the set includes a unified embedding-generation model (ULME) that does not rely on external retrieval but instead learns embeddings jointly with generation through learnable slots and a predictive loss, achieving consistent gains on embedding benchmarks while preserving generation quality [P2]. This contrasts with the retrieval-centric designs of P1 and P3 and suggests an alternative route to improving downstream tasks without explicit retrieval.

Finally, the Indonesian nutrition QA benchmark provides a new evaluation arena for RAG systems in a low-resource clinical domain. While the benchmark itself does not propose a new RAG method, it reports that a RAG baseline underperforms zero- and few-shot prompting by large margins, underscoring the need for stronger retrieval-augmented designs in multilingual clinical QA [P4].

4 theme sentence(s) checked against the paper summaries.

At a glance

[P1] From Structured Records to Patient Narratives: Preserving Clinical Alignment in LLM-Generated Health Summaries · relevance 0.32
Constraining LLMs with evidence grounding and retrieval improves clinical alignment of patient-facing health summaries but makes them harder to read, and the study is limited to synthetic menstrual-health data without human validation.

[P2] ULME: Predictive Coupling for Mutually Reinforcing Embedding and Language Modeling · relevance 0.30
ULME is a post-training framework that unifies text embedding and generation in a single causal decoder-only LLM by inserting learnable slots and coupling a contrastive, language-modeling, and stop-gradient predictive loss, yielding consistent gains on embedding benchmarks while matching generation performance.

[P3] Automated Enrichment Analysis with GraphRAG: Efficient Multi-Database ORA Extraction and Biological Interpretation · relevance 0.30
GBOR-RAG unifies multi-database ORA on a term-gene graph, retrieves compact evidence subgraphs, and uses an LLM to produce more faithful biological interpretations while speeding up queries.

[P4] IndoNutriQA: An Indonesian Clinical Nutrition Question Answering Benchmark · relevance 0.25
IndoNutriQA is a 911-pair Indonesian clinical nutrition QA benchmark that enables zero- and few-shot LLM evaluation, with top 5-shot models reaching ~55 F1 but limited by its modest size and fixed question set.

[P1] From Structured Records to Patient Narratives: Preserving Clinical Alignment in LLM-Generated Health Summaries

Bayram Ayadi, Florian Geissler, Johanna Schmidhuber · arXiv 2610.31467v1 · https://arxiv.org/abs/2610.31467v1 · relevance 0.32

TL;DR: Constraining LLMs with evidence grounding and retrieval improves clinical alignment of patient-facing health summaries but makes them harder to read, and the study is limited to synthetic menstrual-health data without human validation.

Problem
Translating longitudinal physiological records from wearables into patient-accessible narratives risks semantic drift that can distort clinically precise meaning [1]. No prior work has combined factuality verification, uncertainty calibration, and longitudinal summarization for structured-to-narrative, patient-facing summarization of wearable health data [2].

Method
The authors build a semi-synthetic menstrual-health dataset of 500 single-day records generated by a causal DAG and assign uncertainty meta-data [2]. Summaries are produced under three regimes--Base (minimal control), Constrained (explicit clinical communication constraints in the prompt), and RAG-Constrained (constraints plus retrieval of lay-term definitions from the README dataset of >50 000 medical-term pairs) [2]. Three LLMs (GPT-4o, Phi-4-mini, and Qwen2.5-14B) are evaluated, yielding 4 500 generated summaries (500 records × 3 models × 3 settings) [4].

Key results
- Faithfulness violation (Mfaith) improves from a pooled mean of 0.242 (Base) to 0.188 (Constrained) and 0.178 (RAG-Constrained); GPT-4o further improves with retrieval (0.186 -> 0.152) while Qwen2.5-14B is essentially unchanged (0.165 -> 0.162) and Phi-4-mini slightly worsens (0.212 -> 0.220) [4]. Across models the lowest pooled Mfaith is 0.172 for Qwen2.5-14B, followed by 0.201 for GPT-4o and 0.235 for Phi-4-mini [4].
- Epistemic misalignment (Mepist) decreases monotonically: pooled means 0.297 (Base), 0.263 (Constrained), 0.248 (RAG-Constrained); GPT-4o attains the lowest final value of 0.237 [4].
- Omission drift shows model-specific trends: GPT-4o drops from 0.219 (Base) to 0.130 (RAG-Constrained) with a median of 0.0; Qwen2.5-14B is best under Constrained (0.147) but rises slightly with retrieval (0.167) while staying below its Base (0.176); Phi-4-mini worsens monotonically from 0.264 to 0.286 [4].
- Urgency misalignment (Murg) rises from Base to Constrained (e.g., GPT-4o 0.170 -> 0.227) then falls back toward Base after retrieval [4].
- Readability declines: constrained and RAG-Constrained outputs are harder to read, less lexically diverse, and contain more medical jargon [5][6].

Limitations
The study relies entirely on automatic metrics (NLI, named-entity recognizer, single LLM judge) with no human validation of faithfulness or urgency judgments [7]. Only a synthetic DAG-generated dataset was used, without clinician review of realism, limiting generalizability to real-world self-tracked data [7]. Retrieval is confined to lay-term definitions and does not provide additional clinical evidence, so findings may not extend to other RAG setups [7]. No external baseline system is compared, and the experiments focus solely on menstrual health, restricting applicability to other physiological domains or diverse

populations [7].

Sources

[1] arXiv 2610.31467v1.pdf, arXiv abstract

[2] arXiv 2610.31467v1.pdf, page 3

[4] arXiv 2610.31467v1.pdf, page 6

[5] arXiv 2610.31467v1.pdf, page 8

[6] arXiv 2610.31467v1.pdf, page 8

[7] arXiv 2610.31467v1.pdf, page 9

[P2] ULME: Predictive Coupling for Mutually Reinforcing Embedding and Language Modeling

Yang Liu, Zhiqiang Yuan, Shu-Dong Huang, Jiancheng Lv et al. · arXiv 2610.30705v1 · https://arxiv.org/abs/2610.30705v1 · relevance 0.30

TL;DR: ULME is a post-training framework that unifies text embedding and generation in a single causal decoder-only LLM by inserting learnable slots and coupling a contrastive, language-modeling, and stop-gradient predictive loss, yielding consistent gains on embedding benchmarks while matching generation performance.

Problem
Current large language models provide embedding and generation via separate specialized models or only partially share parameters, so the two capabilities do not shape each other and remain merely co-existing [1]. Existing unified models either switch attention regimes or optimize parallel objectives without mutual supervision, limiting true unification [1].

Method
ULME inserts a chain of learnable slot tokens between the query and answer in a causal decoder-only LLM, designating slot 0 for a semantic embedding and slot N-1 for a reasoning embedding [2][3]. Three objectives share all parameters: a contrastive loss (L_sem) anchors the semantic slot to document representations, a language-modeling loss (L_gen) supervises answer generation, and a stop-gradient predictive loss (L_pred) aligns the reasoning slot with a pooled summary of the answer-side hidden states, thereby letting generation shape the embedding without interference [2][3]. The predictive coupling is a one-sided distillation from answer-side latents to query-side latents, enforced by an InfoNCE loss with stop-gradient on the answer side [2].

Key results
Across Qwen2.5 and Qwen3 families (0.5 B-8 B parameters), a single ULME checkpoint improves the MTEB overall score by 3.3-4.1 points over the strongest open-source baseline and tightens BRIGHT and AdvBench-IR simultaneously [1][4]. On the Qwen3-8 B backbone, ULME reaches an overall MTEB score of 66.0, a gain of 9.2 points over LLM2Vec and 4.1 points over LLM2VEC-GEN on the same backbone [4]. Category-wide lifts on Qwen3-8 B include Retrieval +9.6, PairClassification +9.0, Summarization +9.0, STS +9.6, and Clustering +9.4, with other categories improving by 4-7 points [4]. Generation performance on standard accuracy benchmarks (MMLU, GSM8K, MMLU-Pro, HumanEval) remains on par with a generation-only fine-tune of the same backbone, as reported in the abstract [1]. Ablations show that removing the predictive loss reduces performance by 3.6 points, and using mean-pooled answer representations as the predictive target yields the best embedding results, outperforming last-token pooling by up to 1.9 points on MTEB-lite, while generation metrics stay unchanged [5].

Limitations
The generation-side training in ULME uses only a small collection of publicly available reasoning corpora [7].

Sources

[1] arXiv 2610.30705v1.pdf, arXiv abstract

[2] arXiv 2610.30705v1.pdf, page 2

[3] arXiv 2610.30705v1.pdf, page 3

[4] arXiv 2610.30705v1.pdf, page 5

[5] arXiv 2610.30705v1.pdf, page 7

[7] arXiv 2610.30705v1.pdf, page 12

[P3] Automated Enrichment Analysis with GraphRAG: Efficient Multi-Database ORA Extraction and Biological Interpretation

Guangcan Li, Shuyuan Zheng, Jiayao Liu, Shigeto Seno et al. · arXiv 2610.30710v1 · https://arxiv.org/abs/2610.30710v1 · relevance 0.30

TL;DR: GBOR-RAG unifies multi-database ORA on a term-gene graph, retrieves compact evidence subgraphs, and uses an LLM to produce more faithful biological interpretations while speeding up queries.

Problem
Over-representation analysis (ORA) results are often split across many gene-set databases and presented as flat term lists that hide the gene-level connections among related enrichment signals, making interpretation fragmented and opaque[1].

Method
The authors introduce Graph-Based Over-Representation Retrieval-Augmented Generation (GBOR-RAG), which (i) builds a unified term-gene knowledge graph from heterogeneous databases, (ii) scores terms with graph-based ORA, (iii) extracts a compact, connected evidence subgraph via prize-collecting retrieval, and (iv) conditions a large language model (LLM) on this subgraph to generate textual interpretations[1][2][3]. They also define a claim-level faithfulness metric to evaluate whether generated claims are supported by the retrieved evidence[1][2], and present an advanced variant that adds a PubMed citation layer to enrich the graph[2].

Key results
- Graph-based ORA reproduces conventional ORA with a mean Pearson correlation of 0.997 while cutting query time by a median 37× (up to 48×)[1][3].
- Retrieved evidence subgraphs contain on average 36.8 nodes and achieve the highest overall integration score of 4.12[1][3].
- GBOR-RAG attains the highest claim-level faithfulness score of 0.90 among compared systems[1][3].
- Adding the PubMed literature layer raises overall interpretation quality from 4.17 to 4.75[1][3].
- In a 42-query benchmark, GBOR-RAG outperforms a top-k pooling baseline in integration, bridging, and overall usefulness ($p < 10?^{3}$) and matches top-k pooling in relevance under strong compression, though it trails Enrichr-KG in relevance[6].
- Compared to top-k pooling, GBOR-RAG produces a single connected subgraph rather than fragmented ones[6].

Limitations
The evaluation relies on an LLM judge calibrated on a small set of human-rated subgraphs, and the reported faithfulness metric measures consistency with retrieved evidence rather than true biological correctness; a larger expert study is left for future work[7].

Sources

[1] arXiv 2610.30710v1.pdf, arXiv abstract

[2] arXiv 2610.30710v1.pdf, page 1

[3] arXiv 2610.30710v1.pdf, page 2

[6] arXiv 2610.30710v1.pdf, page 5

[7] arXiv 2610.30710v1.pdf, page 7

[P4] IndoNutriQA: An Indonesian Clinical Nutrition Question Answering Benchmark

Tirana Noor Fatyanosa, Ummu Ditya Erliana, Mahendra Data, Muhammad Nurwegiono et al. · arXiv 2610.31256v1 · https://arxiv.org/abs/2610.31256v1 · relevance 0.25

TL;DR: IndoNutriQA is a 911-pair Indonesian clinical nutrition QA benchmark that enables zero- and few-shot LLM evaluation, with top 5-shot models reaching ~55 F1 but limited by its modest size and fixed question set.

Problem
Progress in clinical question answering for low-resource languages such as Indonesian is hampered by the lack of public domain-specific data [1]. The IndoNutriQA dataset focuses on eight NCP/PES reasoning steps and does not cover topics beyond these steps [6].

Method
The authors built IndoNutriQA by collecting 120 real clinical-nutrition reports from multiple hospitals, anonymizing them, and annotating anthropometric and diagnostic metadata [2]. Each report was answered by a single nutritionist to eight standardized NCP/PES questions, yielding 911 free-form answer pairs (median 30 words) [2]. The dataset supports generative, extractive, and retrieval-augmented QA and is evaluated via fine-tuned IndoBERT variants, mT5, RAG, and eight contemporary LLMs in zero- and five-shot settings [1,3]. Evaluation includes 5-fold cross-validation, learning-curve analysis, bootstrap 95 % confidence intervals, and semantic as well as rubric-based metrics [4].

Key results
- The best fine-tuned extractive baseline (Rifky_IndoBERT_QA) achieved F1 = 36.34 (zero-shot) and cross-validated F1 = 32.62 ± 1.29 [4].
- In five-shot prompting, Llama-3.1-8B-Instruct reached F1 = 55.55, MedGemma-4B-IT F1 = 54.28, and GPT-4o F1 = 50.33, all statistically tied and surpassing the fine-tuned baseline by > 19 points [4].
- Zero-shot, MedGemma-4B-IT scored F1 = 42.60, outperforming GPT-4o-mini (36.06) and other multilingual models; its five-shot BLEU was 30.07, matching Llama-3.1-8B-Instruct [4].
- Human validation found 6-10 % of reference answers contestable, and expert re-scoring of 50 samples yielded quadratic-weighted ? = 0.780 (factuality), 0.752 (completeness), but only 0.160 (clinical safety) [7].

Limitations
The benchmark contains only 911 QA pairs from 120 reports, limiting statistical power and rare-case coverage; the authors therefore frame it as an evaluation resource rather than a large training corpus [6]. The fixed set of eight questions constrains scope to NCP/PES steps and does not assess open-ended dialogue or other clinical domains [6]. Each (report, question) pair has a single author, preventing conventional inter-annotator ? calculation and mixing author effects with report effects [2]. Domain and population skew (e.g., pediatric and respiratory cases dominate) and the exclusive use of Indonesian language restrict generalizability to other settings and languages [6,7].

Sources

[1] arXiv 2610.31256v1.pdf, arXiv abstract

[2] arXiv 2610.31256v1.pdf, page 4

[3] arXiv 2610.31256v1.pdf, page 5

[4] arXiv 2610.31256v1.pdf, page 8

[6] arXiv 2610.31256v1.pdf, page 10

[7] arXiv 2610.31256v1.pdf, page 11